

Overview of Linear regression

Subject: Linear regression

$$\hat{y} = X\beta + \varepsilon$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

1.

Errors-in-variables Errors-in-variables models (or "measurement error models") extend the traditional linear regression model to allow the predictor variables X to be observed with error. This error causes standard estimators of β to become biased. Generally, the form of bias is an attenuation, meaning that the effects are biased toward zero.

2.

$y_i \approx \sum_{j=0}^m \beta_j \times x_{ji} = \beta \rightarrow \cdot x_i \rightarrow \{\displaystyle y_i \approx \sum_{j=0}^m \beta_j \times x_{ji} = \vec{\beta} \cdot \vec{x_i}\}$. In the least-squares setting, the optimum parameter vector is defined as such that minimizes the sum of mean squared loss:

3.

Further reading Pedhazur, Elazar J (1982). Multiple regression in behavioral research: Explanation and prediction (2nd ed.). New York: Holt, Rinehart and Winston. ISBN 978-0-03-041760-3. Mathieu Rouaud, 2013: Probability, Statistics and Estimation Chapter 2: Linear Regression, Linear Regression with Error Bars and Nonlinear Regression. National Physical Laboratory (1961). "Chapter 1: Linear Equations and Matrices: Direct Methods". Modern Computing Methods. Notes on Applied Science. Vol. 16 (2nd ed.). Her Majesty's Stationery Office.

4.

which has an interpretation as the expected change in y' $\{\displaystyle y'\}$ when all x_j' $\{\displaystyle x_j'\}$ in the strongly correlated group increase by $(1/q)$ $\{\displaystyle (1/q)\}$ th of a unit at the same time with variables outside the group held constant. With strong positive correlations and in standardized units, variables in the group are approximately equal, so they are likely to increase at the same time and in similar amount. Thus, the average group effect ξ_A $\{\displaystyle \xi_A\}$ is a meaningful effect. It can be accurately estimated by its minimum-variance unbiased linear estimator $\hat{\xi}_A = 1/q (\hat{\beta}_1' + \hat{\beta}_2' + \dots + \hat{\beta}_q')$ $\{\text{style } \hat{\xi}_A = \frac{1}{q} (\hat{\beta}_1' + \hat{\beta}_2' + \dots + \hat{\beta}_q')\}$, even when individually none of the β_j' $\{\displaystyle \beta_j'\}$ can be accurately estimated by $\hat{\beta}_j'$ $\{\displaystyle \hat{\beta}_j'\}$. Not all group effects are meaningful or can be accurately estimated. For example, β_1' $\{\displaystyle \beta_1'\}$ is a special group effect with weights $w_1 = 1$ $\{\displaystyle w_1=1\}$ and $w_j = 0$

$w_j = 0$ for $j \neq 1$, but it cannot be accurately estimated by $\hat{\beta}_1$. It is also not a meaningful effect. In general, for a group of q strongly correlated predictor variables in an APC arrangement in the standardized model, group effects whose weight vectors $\mathbf{w}$ are at or near the centre of the simplex $\sum_{j=1}^q w_j = 1$ ($w_j \geq 0$) are meaningful and can be accurately estimated by their minimum-variance unbiased linear estimators. Effects with weight vectors far away from the centre are not meaningful as such weight vectors represent simultaneous changes of the variables that violate the strong positive correlations of the standardized variables in an APC arrangement. As such, they are not probable. These effects also cannot be accurately estimated. Applications of the group effects include (1) estimation and inference for meaningful group effects on the response variable, (2) testing for "group significance" of the q variables via testing $H_0: \xi_A = 0$ versus $H_1: \xi_A \neq 0$, and (3) characterizing the region of the predictor variable space over which predictions by the least squares estimated model are accurate. A group effect of the original variables $\{x_1, x_2, \dots, x_q\}$ can be expressed as a constant times a group effect of the standardized variables $\{x_1', x_2', \dots, x_q'\}$. The former is meaningful when the latter is. Thus meaningful group effects of the original variables can be found through meaningful group effects of the standardized variables.

5.

$$\xi'(\mathbf{w}) = w_1 \beta_1' + w_2 \beta_2' + \dots + w_q \beta_q', \quad \xi(\mathbf{w}) = w_1 \beta_1 + w_2 \beta_2 + \dots + w_q \beta_q$$

6.

parameter β_j of predictor variable x_j represents the individual effect of x_j . It has an interpretation as the expected change in the response variable y when x_j increases by one unit with other predictor variables held constant. When x_j is strongly correlated with other predictor variables, it is improbable that x_j can increase by one unit with other variables held constant. In this case, the interpretation of β_j becomes problematic as it is based on an improbable condition, and the effect of x_j cannot be evaluated in isolation. For a group of predictor variables, say, $\{x_1, x_2, \dots, x_q\}$, a group effect $\xi(\mathbf{w})$ is defined as a linear combination of their parameters

7.

Bayesian linear regression applies the framework of Bayesian statistics to linear regression. (See also Bayesian multivariate linear regression.) In particular, the regression coefficients β are assumed to be random variables with a specified prior distribution. The prior distribution can bias the solutions for the regression coefficients, in a way similar to (but more general than) ridge regression or lasso regression. In addition, the Bayesian estimation process produces not a single point estimate for the "best" values of the regression coefficients but an entire posterior distribution, completely describing the uncertainty surrounding the quantity. This can be used to estimate the "best" coefficients using the mean, mode, median, any quantile (see quantile regression), or any other function of the posterior distribution.

Quantile regression focuses on the conditional quantiles of y given X rather than the conditional mean of y given X . Linear quantile regression models a particular conditional quantile, for example the conditional median, as a linear function $\beta^T x$ of the predictors. Mixed models are widely used to analyze linear regression relationships involving dependent data when the dependencies have a known structure. Common applications of mixed models include analysis of data involving repeated measurements, such as longitudinal data, or data obtained from cluster sampling. They are generally fit as parametric models, using maximum likelihood or Bayesian estimation. In the case where the errors are modeled as normal random variables, there is a close connection between mixed models and generalized least squares. Fixed effects estimation is an alternative approach to analyzing this type of data. Principal component regression (PCR) is used when the number of predictor variables is large, or when strong correlations exist among the predictor variables. This two-stage procedure first reduces the predictor variables using principal component analysis, and then uses the reduced variables in an OLS regression fit. While it often works well in practice, there is no general theoretical reason that the most informative linear function of the predictor variables should lie among the dominant principal components of the multivariate distribution of the predictor variables. The partial least squares regression is the extension of the PCR method which does not suffer from the mentioned deficiency. Least-angle regression is an estimation procedure for linear regression models that was developed to handle high-dimensional covariate vectors, potentially with more covariates than observations. The Theil–Sen estimator is a simple robust estimation technique that chooses the slope of the fit line to be the median of the slopes of the lines through pairs of sample points. It has similar statistical efficiency properties to simple linear regression but is much less sensitive to outliers. Other robust estimation techniques, including the α -trimmed mean approach, and L-, M-, S-, and R-estimators have been introduced.